Chapai - A Print Management System

Muhammad Ahmed Qazi

September 2026

Abstract

This article specifies the software requirements for a smart printing management system to be deployed in a university campus environment.

1 Introduction

Traditional paper document printing architectures on university campuses represent a significant logistical bottleneck, operational inefficiency, and economic vulnerability.

To overcome these dificiencies, "Chapai" introduces an automated, transactionally secure, hardware-aware, and priority-driven smart campus printing management network.

1.1 User Requirements

Following are the requirements that our system fulfils:

1. **Authentication:** A user should be able to create an account and login to it.
2. **Print:** A user (student or faculty) must be able to send a document to be printed with the desired configuration to our printing kiosks.
3. **Manage credit:** A user should be able to do the following:
   - See the balance they have in their account.
   - Add some balance to their account.
   - See their transaction history.
4. **Tracking:** A user should be able to track the status of their document whether it has been queued, being printed, or ready for collection.

2 Backend and Front-End Tools

Our system uses the following tools to lay the foundation of our system, i.e. *back-end tools*:

- **Database Management System:** *Oracle Database 21c Express Edition (XE)* with *Oracle SQL Developer as GUI.*
- **Backend Architecture:** *NestJS Framework*, an enterprise-grade TypeScript framework built atop Node.js, implementing a strict modular structure and managing standard Oracle client database drivers via optimised persistence layers.
- **Containerization Subsystem:** Podman Container Engine, utilised to sandbox the Oracle databse environment, ensuring complete reproducibility across developer workstations and preventing operating system dependency pollution.

The following tools are used to build the interfaces for our system, i.e. *front-end tools*:

- **Web Interface:** *Angular Framework.*
- **Mobile Interface:** *Ionic Framework* combined with *Angular 17.*

3 Software Attributes

Our print management system has the following software attributes:

- Functionality: print job submission, authentication, account balance and transactions.
- Usability: the system has two distinct interface for a user and an operator which are both designed to accommodate the technical ability of the user.
- Portability: this system is both deployed as a desktop website and an Android mobile application.

4 Conclusion

The Chapai platform successfully demonstrates a highly functional, enterprise-grade, three-tier application designed to modernize and streamline printing operations across the NED University campus. By replacing inefficient, manual printing queues and unverified cash transactions with a centralized digital ledger and intelligent hardware routing, the system addresses core logistical bottlenecks for both students and administration.

The platform heavily leverages advanced capabilities inherent to the Oracle 21c ecosystem. Complex business operations, such as simultaneous balance deductions and ledger updates, are securely encapsulated inside PL/SQL stored procedures, guaranteeing strict transactional atomicity. Database triggers autonomously maintain referential integrity across unlinked relational bridges, while highly optimized SQL views dynamically flatten complex user hierarchies to provide real-time analytical reporting. By enforcing data isolation at the database layer and controlling access through a robust NestJS API, the architecture ensures a highly secure and highly available production environment.